ASR-Aware Unified Fusion of Three Speech Enhancers for Noise Event Removal on Vaani

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***Abstract*—We remove noise events from Indic field speech so that both the waveform (SI-SDR on synthetic clips) and the recognisability of the speech (word error rate reduction, ΔWER, of the mandated SraVaani-1.0 ASR) improve. Our system processes every clip identically: three public pretrained enhancers of different families (MossFormerGAN, a 104M band-split RNN, TF-Locoformer) are fine-tuned *through the frozen scoring ASR* with an encoder-distance loss, a TDT transcript likelihood and a deletion-weighted minimum-word-error (MWER) loss, using a long single-recording training stream that matches the test clip length; a 0.35M-parameter network fuses their estimates per time-frequency cell. The final entry scores Combined 19.90 (SI-SDR 15.79 dB; ΔWER $+11.66$ pp synthetic, $+0.54$ pp natural) and passed the organisers' ASR audit. A diagnosis of where words are lost shows that the remaining synthetic errors are deletions under loud events and that natural-clip errors are mostly not caused by noise.**

I. Training Data and Preprocessing

Vaani validation set. The organisers' validation release provides 695 synthetic pairs (clean reference and noisy mixture, 4.5 s median) and 1,806 natural field recordings with human transcripts. We split validation speakers by a fixed hash (70/30): 486 pairs train the *held-out-protocol* models and 209 pairs from unseen speakers are kept for honest scoring and as fusion training inputs. Shipped MossFormerGAN and BSRNN models are retrained on all 695 pairs. The 555-clip speaker-disjoint half of the natural clips is used only to measure natural ΔWER.

Noise events (augmentation only). The organisers' mixing is additive, so noisy minus clean recovers each interferer exactly; cutting it at the annotated event spans gives 2,213 noise events. They are used only to synthesise training mixtures; nothing at inference reads, matches or subtracts them.

Augmentation pipeline. Each crop (2 s for MossFormerGAN, 4 s otherwise) is, with probability 0.5, a real pair, else a clean reference re-mixed with 1–3 random events at an SNR drawn from the real pairs' SNRs; mixtures above full scale are normalised. A *long stream* adds one micro-batch per step of 12–18 s single-speaker windows from 1,450 recordings, with events pasted as the organisers' generator does (Poisson 0.5 events/s, gain log-normal around 0.93, $\sigma = 0.8$) and the bed level drawn from the references' RMS; it trains the ASR losses only.

External data. ARTPARK-IISc/Vaani-transcription-part [1] supplies the long beds and part of the fusion beds; IndicVoices-R [2] supplies fusion beds. Every competition clip and every recording behind a competition clip was excluded, beds were screened for quiet pauses, benign transcript tags and no clipping, and levels follow the organiser references. SraVaani greedy tokens of each clean bed, decoded once at full context, serve as transcript teachers. No test audio is used for training.

TABLE I
Data used. Only the first two rows are organiser data; nothing from the test set is used for training.

Set	Source	Clips	Hours	Use
Synthetic pairs	Vaani validation	695	1.0	members (486), held-out (209)
Natural clips	Vaani validation	1,806	2.5	natural ΔWER (555)
Noise events	noisy $-$ clean of the pairs	2,213	0.4	augmentation only
Long beds	Vaani transcription release	1,450	5.7	long ASR stream
fresh2	IndicVoices-R, Vaani beds + events	5,000	19.8	fusion training (2,500)
valset	held-out beds + events	1,200	4.2	fusion training, evaluation
valset_long	unseen Vaani speakers + events	360	1.3	evaluation (test length)

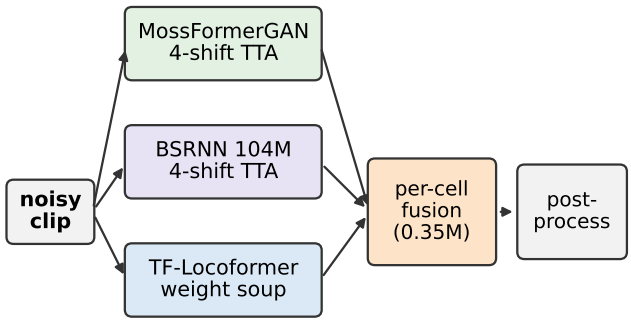


Fig. 1. The unified pipeline. Every clip takes the same path; the fusion predicts per-cell member weights and a bounded readmission of the input.

Simulated sets for the fusion. Table I summarises all data (long beds: 1,291 speakers, 29 languages; noise events: 23 min in 7 categories). *fresh2*: 5,000 test-shaped single-speaker mixtures (8–18 s, 0.65 events/s) on beds unseen by every held-out-protocol member (2,500 used); *valset*: 1,200 Phase-2-shaped clips on held-out speakers; *valset_long*: 360 test-length clips from unseen speakers for evaluation.

II. Proposed Method

Members. Fig. 1 shows the system and Table II lists the pretrained start points and roles. All three are fine-tuned with

$$\mathcal{L} = -\text{SI-SDR}(\hat{s}, s) + 30\, d_{\text{enc}}(\hat{s}, s) + 2.5\, \mathcal{L}_{\text{NLL}}(\hat{s}) + \lambda\, \mathcal{L}_{\text{MWER}}(\hat{s}), \quad (1)$$

TABLE II
PRETRAINED MODELS AND THEIR ROLES.

Model (source, version)	Params	Role
MossFormerGAN_SE_16K (ClearVoice [6])	3.1M	member 1, 4-shift TTA
BSRNN large-double, VCTK+DNS+WHAMR! (ESPnet [7], `wyz/..._large_double`)	104M	member 2, 4-shift TTA
TF-Locoformer M, DNS-2020 (MERL [8])	15M	member 3, weight soup
SraVaani-1.0 (rev. `f5dd535`)	–	frozen loss, transcripts

TABLE III
TRAINING STAGES OF THE SHIPPED MEMBERS (STEPS; CROP; LR; LOSSES; DATA). ENC = ENCODER DISTANCE, NLL = TRANSCRIPT LIKELIHOOD, LS = LONG SINGLE-RECORDING STREAM; "ALL" = 695 PAIRS, "H/O" = 486-PAIR HELD-OUT PROTOCOL.

Member	Stage	Steps	LR	Losses	Pairs
MossFormerGAN	pre-train	6k+34k	1e-4	SI-SDR+L1	h/o
(2 s crops, bs 4)	encoder	4k+8k	5e-5	+Enc	all
	transcript	4k	5e-5	+NLL	all
	long stream	4k	5e-5	+LS	all
BSRNN-104M	encoder	30k	1e-4	SI-SDR+Enc	all
(4 s crops, bs 8)	transcript	6k	5e-5	+NLL	all
	long stream	6k	5e-5	+LS	all
TF-Locoformer	long stream	20k	1e-4	all + LS	h/o
(4 s crops, bs 4)	3 fine-tunes	3×4k	5e-5	+MWER/–/loud	h/o
Fusion	–	3k	5e-4	all four	held-out

where d_{enc} is the mean per-frame cosine distance between SraVaani encoder states of the estimate and of clean speech, computed through a differentiable copy of the ASR front end; $\mathcal{L}_{\text{NLL}}$ is the TDT [3] negative log-likelihood of the clean speech's teacher tokens on the estimate's lattice (a custom forward recursion, with crop edges handled by letting paths start and end inside a margin); and $\mathcal{L}_{\text{MWER}}$ draws $N{=}4$ ancestral samples from the TDT decoder per window of 8 s or more and applies REINFORCE with a mean baseline to their word cost against the teacher transcript, a deletion costing 2 words [4]. The ASR is frozen; gradients reach only the enhancer. MossFormerGAN keeps its compressed-spectrum L1 term. Optimisation uses AdamW, warm-up then cosine decay, gradient clipping at 5 and an EMA of the weights (0.999–0.9995), in stages: noise-removal pre-training, then the encoder loss, the transcript loss and finally the long stream (MossFormerGAN 9 stages, BSRNN 6; exact configurations are in the repository). TF-Locoformer starts from MERL's checkpoint with all losses and the long stream for 20k steps; three 4k-step continuations (with MWER $\lambda{=}3$, without, and with events 6 dB louder) are averaged into one model, a weight soup [5] that costs nothing at inference.

Fusion. The members' outputs are least-squares scaled onto the input and transformed (512-point STFT, hop 128). From the members' and the input's log powers, each member's phase and level relative to a fixed mix $B = \sum_m w_m^0 X_m$, and a frequency position, eight dilated residual convolution blocks (64 channels) predict per cell softmax weights w_m (initialised at $w^0 = (0.2, 0.35, 0.45)$) and a complex readmission r of the noisy cell bounded by $|r| \le 0.2$ and scaled by how far the mix moved that cell:

$$\hat{S} = \textstyle\sum_m w_m X_m + r\, a\, Y, \quad a = \min\big(1, |Y - B|/|Y|\big). \quad (2)$$

The fusion trains for 3,000 steps on out-of-sample member outputs (held-out-protocol members on fresh2, the 209 held-out pairs and valset) with the same four losses (MWER $\lambda{=}3$), back-propagated through the shipped post-processing.

TABLE IV
PHASE 2 BOARD, ONE CHANGE PER ROW (ΔWER IN PP). ROWS ABOVE THE RULE PREDATE THE ORGANISERS' UNIFIED-PIPELINE RULE AND ROUTED NATURAL CLIPS THROUGH A LIGHTER PATH; THE ROWS BELOW, AND THE FINAL SYSTEM, PROCESS EVERY CLIP IDENTICALLY.

System	Comb.	SI-SDR	syn.	nat.
Initial 3-model blend	16.15	13.63	7.11	0.35
+ ASR-encoder loss	16.51	13.76	7.58	0.47
+ BSRNN family	16.97	14.16	7.83	0.43
+ transcript NLL, 104M BSRNN	17.48	14.34	8.82	0.46
+ learned per-cell fusion	17.79	14.49	9.29	0.48
+ TF-Locoformer	19.16	15.42	10.60	0.50
Unified pipeline, long-clip stream	19.30	15.48	11.01	0.43
+ fusion readmission, test-length data	19.68	15.75	11.30	0.46
Three members only	19.70	15.75	11.30	0.49
All three with the long stream	19.81	15.77	11.49	0.52
+ MWER on TF-Locoformer	19.83	15.78	11.52	0.53
+ TF-Locoformer weight soup	19.86	15.80	11.59	0.52
+ MWER in the fusion (final)	**19.90**	**15.79**	**11.66**	**0.54**

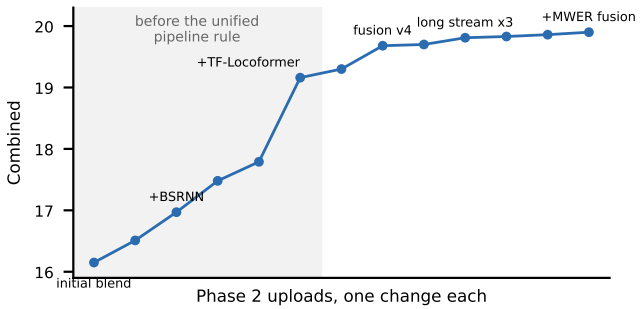


Fig. 2. Phase 2 board score per upload (Table IV); the largest step is the new TF-Locoformer family.

Inference. MossFormerGAN and BSRNN average four input delays (0, 41, 83, 124 samples); long clips are decoded in overlapping windows (TF-Locoformer 8 s/6 s hop, each window gain-matched, BSRNN 20 s/15 s). Post-processing is identical for every clip: the output is kept where the model changed the audio (smoothed residual above 1.5% of its maximum, dilated 150 ms, 10 ms cross-fades), a -50 dB silence gate and a 0.99 peak guard; on test clips the shipped audio differs from the pure model output by -41 dB (median). There is no clip classifier, no branch by clip type, no pass-through of whole clips and no template, dictionary or correlation matching.

III. RESULTS AND CONCLUSION

The metric is Combined = SI-SDR [9] (synthetic clips) $+ 100 \cdot \Delta$WER (all 3,178 clips, pooled). Table IV and Fig. 2 show how the system was built on the Phase 2 board; each row adds one change.

Validity. Changes were screened on the 209 held-out pairs, the 360 test-length clips and the 555 natural clips with paired clip-bootstrap intervals before any upload. Table V isolates

TABLE V

EACH COMPONENT MEASURED LOCALLY AGAINST THE SAME SYSTEM WITHOUT IT (PAIRED CLIP BOOTSTRAP, 95% CI; SIM. = 1,200 SIMULATED CLIPS, 33.8K WORDS; LONG = 360 TEST-LENGTH CLIPS; H/O = 209 HELD-OUT PAIRS), AND ITS BOARD EFFECT WHERE IT WAS UPLOADED ALONE. †LOCAL ON MOSSFORMERGAN, BOARD ON AN MP-SENET MEMBER.

Component	Local effect	Board
Encoder loss†	+0.30 pp [+0.06, +0.53] sim.	+0.17
Enc. + transcript loss (BSRNN)	+0.61 pp [+0.36, +0.85] sim., +0.05 dB	–
Long single-recording stream	+1.47 pp [+1.18, +1.78] sim., +0.30 dB	–
TF-Locoformer in the mix	+0.79 dB long (351 of 360 clips better)	+1.37
Learned fusion vs fixed mix	+0.40 pp [+0.21, +0.59] sim., +0.06 dB h/o	+0.13 to +0.24
Long stream on all 3 members	+0.33 pp [+0.07, +0.60], +0.23 dB long	+0.11
TF-Locoformer weight soup	+0.085 dB [+0.056, +0.118] h/o	+0.03
MWER in the fusion	+0.26 pp [+0.10, +0.43] long	+0.04

TABLE VI

WHERE WORDS ARE LOST: WORD ERROR (%) BY REGION ON THE 555 NATURAL CLIPS (WORDS TIMED BY ASR TIMESTAMPS, LABELLED BY THE ORGANISERS' EVENT TIMESTAMPS) AND ΔWER (PP) BOUNDS ON THE 360 TEST-LENGTH CLIPS.

Natural clips	Words	Noisy	Fusion
Clear speech (no event)	5,052	25.4	25.6
Under a short event	2,033	26.8	27.0
Under clip-long ambient noise only	749	32.7	32.3
Test-length clips			**ΔWER**
Clean speech (ceiling)	12,730		+7.59
Speech intact, events at −20 dB (oracle)	12,730		+6.54
Ours	12,730		+4.1

each component: every one kept improved words or SI-SDR locally with an interval excluding zero, and each one uploaded alone also improved the board. Local SI-SDR gains transferred at roughly 25–100%, word gains less reliably (one upload resolves synthetic words to about ±0.25 pp), so each upload changed one thing and the final file was chosen on the board.

Key findings. (i) Training through the scoring ASR helped every family: on 33.8k simulated words the encoder and transcript losses gave +0.3 to +0.6 pp at no SI-SDR cost, and long single-recording windows a further +1.5 pp. (ii) Decorrelated families matter more than size: each member alone scores 0.6–1.0 below the blend, and TF-Locoformer, the one new family whose errors correlate least with the others, gave the largest single gain (+1.37). Two members lose 0.08 against three, while eight members gained nothing over three. (iii) Learning the combination per cell beats fixed weights (+0.1 to +0.4). (iv) Where words are lost: on simulated clips 71% of lost words are deletions and 79% lie under event spans, worst for events louder than −10 dB and longer than 1.2 s; an oracle mask on the mixture recovers 95–97% of the clean-speech word ceiling, so the limit is per-cell gain accuracy under loud events. On natural clips (Table VI), clear speech already has 25.4% word error, words under events only 1.4–7 pp more, and 10.5% of reference words are English glosses that no output can match, which caps the natural gain near 1 pp. On test-length synthetic clips, keeping the speech intact matters more than removing every trace of an event: speech intact with events at −20 dB keeps 86% of the clean-speech word gain.

What did not help. Table VII lists the ideas that did not beat the system, including generative restorers and more clean speech.

TABLE VII

IDEAS THAT DID NOT BEAT THE SYSTEM, MEASURED PAIRED AGAINST IT (LOCAL SETS: TEST-LENGTH LONG SET, 209 HELD-OUT PAIRS; DB IS SI-SDR, PP IS ΔWER).

Idea	Result
Generative restorer as a member (UniPASE)	−8.6 dB alone, −3.5 pp held-out
198 h more clean Indic speech (all 4 families)	−0.05 to −0.17 dB held-out
Trained two-stage TF-Locoformer in the fusion	+0.02 dB, −0.02 pp
Speech-intact target (clean + noise at −20 dB)	−4.5 dB, −0.20 pp (member)
ASR encoder features as fusion inputs	−0.05 dB, +0.03 pp
MWER also on BSRNN and MossFormerGAN	−0.07 dB, +0.06 pp
Words-first fusion (ASR losses ×3)	board 19.88 vs 19.90
TF-Locoformer soup retrained on all pairs	board 19.87 vs 19.90

Reproducibility and code. Code, training configurations for all 19 member stages and the fusion, data preparation scripts and the inference script are at https://github.com/XXXX/indoml2026-track2-hydra and the final checkpoints at https://huggingface.co/manishjoshiai/indoml2026-track2-noise-removal. In a freshly built environment the inference script reproduced the submitted audio to within one 16-bit step on every sample. Training ran on NVIDIA A100 80 GB GPUs (one per run, 20–40 GB each); inference needs one GPU.

Limitations. Four-shift averaging in two members makes inference several times slower than a single pass; local word gains under 0.3 pp could not be confirmed on the board, whose single-upload noise is about ±0.25 pp; and the system leaves about 45% of the test-length clean-speech word gain unrecovered, mostly deletions under loud, long events.

Conclusion. Three fine-tuned public enhancers, trained against the scoring ASR's own objectives and fused per cell, reach Combined 19.90 (fourth on the Phase 2 board) with every clip processed identically.

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